

Mathematical Exploration – The Newton-Raphson method

Introduction

Newton's method, or the Newton-Raphson method, is an iterative method in numerical analysis for the approximation of the roots of non-linear, real-valued functions. It is an algorithm that is mainly based on calculus, condensing the first few terms of a Taylor series, which distills repeated linearizations of root-approximations, into a single formula. In spite of its conceptual simplicity, its applications are both complex and highly significant, providing computational methods for the solution and optimization of algebraically unsolvable equations. This investigation shall first elucidate the Method's significance and functions, giving a rationale for studying it. This shall be followed by a mathematical derivation of the Method's iterative algorithm, as well as sample computations comparing solutions to functions that are algebraically solvable and unsolvable. After examining the method's weakness and possibilities for failure, an algorithm for optimization shall be derived and applied to a mathematical problem in the field of civil engineering.

The first published account of Newton's iterative method appeared in chapter 94 of John Wallis' *A Treatise on Algebra* in 1685 (Ypma 531). Yet Newton's earliest explanation of the method is documented in his tract *Analysis of Infinite Series*, which he composed in 1669 but which was not printed until 1704 (Cajori 29). Between the dates of composition and publication, the method became known to Newton's teacher, Isaac Barrow, and to several members of the Royal Society in London with whom Barrows held correspondence (Cajori 29). English mathematician Joseph Raphson published a modified and condensed interpretation in 1690 (Cajori 31). Raphson reformulated Newton's complicated sequence of polynomials into a simpler succession of root-approximations, the efficiency of which allowed Raphson's approach to largely supplant Newton's original in numerical analysis (Cajori 31). As the method's applications have been extended, its reach has also been extended by mathematicians. British mathematician Arthur Cayley investigated, in the late 1870s, the difficulties of generalizing the method to computing the complex roots of polynomials with degree greater than two (Ypma 531). The generalizations of Newton's method into the complex plane, rational functions, and transcendental functions has since been intensely studied.

Rationale

The Newton-Raphson method still constitutes one of the most significant and effective iterative root-approximating methods in numerical analysis. This is because most higher-degree polynomial equations, as well as many sigmoid and transcendental functions, remain algebraically unsolvable. In fact, the Abel-Ruffini theorem provides proof for the impossibility of a general algebraic solution to polynomial equations of degree five or higher (Abel 66). The Norwegian mathematician Niels Henrik Abel provided a full proof in 1823, after Paolo Ruffini advanced an incomplete one in 1799 (van der Waerden 86). Yet the solutions to these types of functions are fundamental to numerous mathematical models and instruments in both the human and the natural sciences. Newton's method, along with its sibling algorithms, provide an iterative method for the approximation of the roots of precisely these algebraically unsolvable, real-valued functions (under broad conditions).

As already indicated above, the significance of Newton's method is profound. Most graphical calculators use Newton's algorithm to calculate roots. Numerous social, macroeconomic, and physical phenomena are modeled with algebraically unsolvable systems, yet whose root and derivative analysis is fundamental to the study and understanding of these phenomena. The Malthusian model of population growth in macroeconomic, for instance, takes the form of a sigmoidal logistic function that, under certain conditions, is algebraically rather difficult to manage (Weisstein). When used for multi-dimensional optimization, Newton's method may be generalized to linear algebra, there facilitating the study of complex wave and light phenomena modeled by intricate, multivariable transcendental functions (Cantrell 93). Yet the study of these elusive of concepts is fundamental to modern particle physics, quantum mechanics, and cosmology (Cantrell 93). Furthermore, most other iterative methods of linear and quadratic approximation are based on Newton's concept, and this family of algorithms provide the only real platform for work with non-linear systems and models in the social and human sciences.

Derivation

Let $f(x)$ be a real-valued ($x \in \mathbb{R}$), differentiable function defined throughout its domain. An arbitrary approximation of the root of $f(x)$ ($x: f(x) = 0$) must be made, defined as x_n . A formula for a more accurate approximation, x_{n+1} , may subsequently be derived through a linearization of $f(x)$ at x_n . This linearization is equivalent to the tangent line to the curve $y = f(x)$ at the point $x = x_n$. The equation of this tangent (in the form $y = mx + b$) is given by:

$$y - f(x_n) = f'(x_n)(x_{n+1} - x_n)$$

where $f'(x_n)$ is the first derivative of $f(x_n)$ at $x = x_n$.

The x-intercept of this tangent line is then used as the better approximation, x_{n+1} . This approximation is computed by solving the equation of the tangent ($x: f(x) = 0$);

$$0 - f(x_n) = f'(x_n)(x_{n+1} - x_n)$$

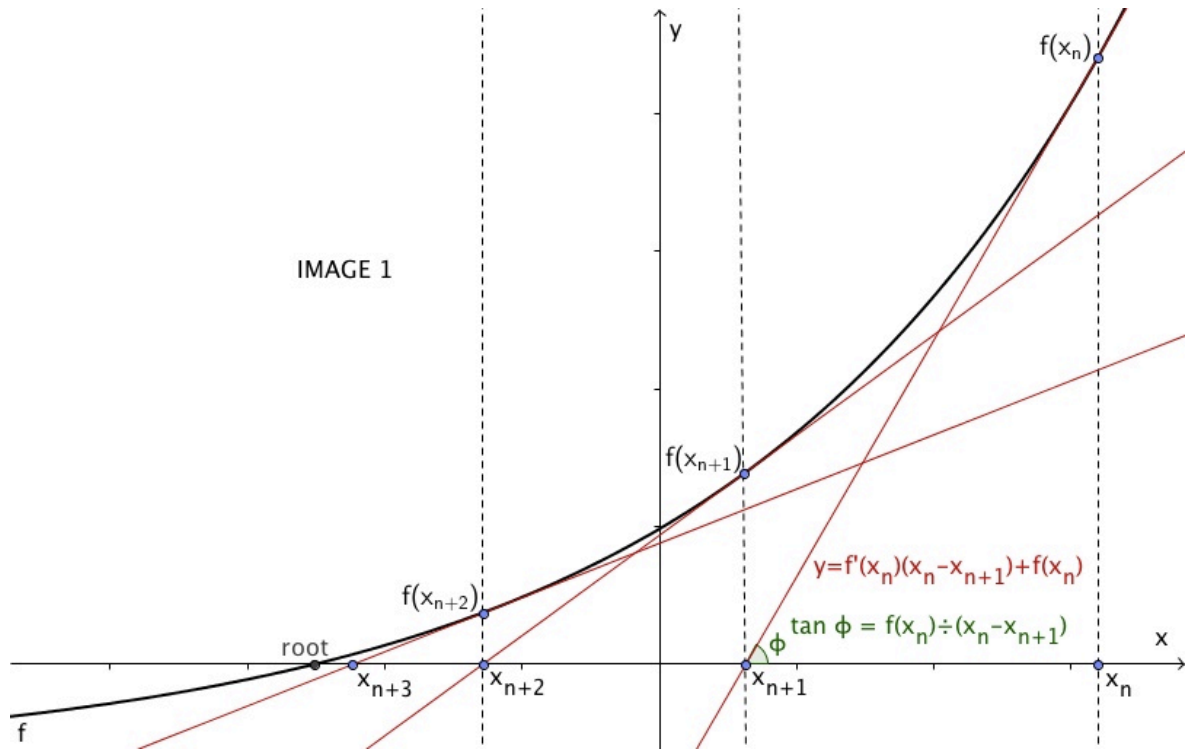
$$-f(x_n) = f'(x_n)x_{n+1} - f'(x_n)x_n$$

$$f'(x_n)x_{n+1} = f'(x_n)x_n - f(x_n)$$

$$x_{n+1} = \frac{f'(x_n)x_n}{f'(x_n)} - \frac{f(x_n)}{f'(x_n)}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

where $f'(x_n) \neq 0$. Thus a formula for a "better" approximation of the function's root may be derived, and subsequently iteratively used to deduce increasingly accurate approximations. See the following for a graphical representation (Image 1) of the process when applied to the function $f(x) = 2^x - 1$. (All graphs and visualizations contained in this investigation have been designed and created independently with the dynamic geometry software GeoGebra.)



An alternative method of derivation involves the angle of interception. Suppose the linearization of $f(x)$ at the approximation $x = x_n$ (point $(x_n, f(x_n))$) intersects the x -axis at x_{n+1} (what is to become the better approximation). Let the angle of interception between this linearization and the x -axis be described by ϕ . In this case,

$$\tan \phi = \frac{\text{rise}}{\text{run}} = \frac{\Delta y}{\Delta x} = \frac{f(x_n) - 0}{x_n - x_{n+1}} = f'(x_n)$$

Solving for x_{n+1} , thus;

$$f'(x_n) = \frac{f(x_n)}{x_n - x_{n+1}}$$

$$x_n - x_{n+1} = \frac{f(x_n)}{f'(x_n)}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

where $f'(x_n) \neq 0$. With this algorithm, an arbitrary initial estimate x_n may be used to iteratively compute increasingly accurate approximations of the functions roots until these approximations converge with the sought root. Convergence succeeds in most cases, provided the conditions for convergence are satisfied and the various failure possibilities (which shall be elaborated on later) are avoided.

In fact, the conditions for successful convergence concern mainly the selection of an appropriate initial estimate. A poor or inappropriate initial approximation makes the algorithm very vulnerable to failure (the ways in which this happens shall be described later) and is a determinant for the iteration's efficiency (Weissstein). The Proof of Quadratic Convergence for Newton's Iterative Method establishes the conditions which an initial approximation must satisfy to successfully converge (Gockenbach). The exact nature of these conditions shall not be elaborated here – their

derivation and explanation involves linear algebra and multivariable calculus of a complexity that is beyond the scope of this investigation. Suffice it to say that the Proof determines that quadratic convergence is assured if: the second derivative of the function under consideration must be finite and the initial estimate should be sufficiently close to the sought root, meaning that linearization across stationary points or asymptotes should be avoided (Weisstein). Thus for the sample computations and problems shown, the assumption has to be made that the initial approximations chosen do in fact satisfy these conditions.

Nevertheless, it should be mentioned that if the conditions for convergence are satisfied, an initial approximation must converge with the sought root (Weisstein). Also, Newton's method is rather efficient; its rate of convergence is typically quadratic, which means that with every iterative step the number of correctly approximated digits roughly doubles (Weisstein).

Sample computation

A root of the function

$$y = f(x) = \frac{1}{2} \tan(x - 2)$$

may be approximated with the Newton-Raphson method and its iterative formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

where $f'(x_n) \neq 0$. It must be noted that an algebraic solution to the here selected function exists; thus a useful comparison can be made between the algebraic and the iterative outcomes. First, the derivative of the function in consideration must be derived. The process of differentiating y in this case can be subdivided into the simpler differentiation of $y = \arctan(x)$ on its own, and then applying the chain rule to deduce the actual derivative formula of the function in consideration.

The derivative of the arctangent function is thus found:

$$y = \arctan x$$

$$\tan y = \tan(\arctan x)$$

$$\tan y = x$$

$$\frac{d}{dx}(\tan y) = \frac{d}{dx}(x)$$

$$(\sec^2 y) \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$\therefore \frac{dy}{dx} = \frac{1}{1 + (\tan y)^2}$$

$$\therefore \frac{dy}{dx} = \frac{1}{1 + x^2}$$

$$\frac{d}{dx}(\arctan x) = \frac{1}{1 + x^2}$$

$$(\sin^2 y + \cos^2 y = 1$$

$$\frac{\sin^2 y}{\cos^2 y} + \frac{\cos^2 y}{\cos^2 y} = \frac{1}{\cos^2 y}$$

$$\tan^2 y + 1 = \sec^2 y)$$

$$(\tan y = x)$$

Subsequently the chain rule is applied to deduce the derivative function for $f(x)$:

$$f(x) = \frac{1}{2} \arctan(x - 2)$$

$$f'(x) = \frac{1}{2} \left(\frac{1}{1 + (x - 2)^2} \right) \cdot 1 = \frac{1}{2 + 2(x - 2)^2}$$

This derived equation may then be inserted into the iterative formula in order to complete the Newton-Raphson algorithm for the function $f(x)$. For

$$y = f(x) = \frac{1}{2} \tan(x - 2)$$

the Newton algorithm, hence, is the following:

$$x_{n+1} = x_n - \frac{\frac{1}{2} \arctan(x_n - 2)}{\frac{1}{2 + 2(x_n - 2)^2}} = x_n - \left(\frac{1}{2} \arctan(x_n - 2) \right) (2 + 2(x_n - 2)^2)$$

To approximate a root in this case, an initial random approximation x_0 at $x = 3.32$ is made. (This value is an arbitrary estimation selected at random for the purposes of demonstration. It is given that this value satisfies the conditions set out by the Proof, and the actual process of selection shall be elaborated on at a later stage.) This initial estimate may then be used to iterate the function's root in the vicinity of $x = 3.32$:

$$x_0 = 3.32$$

$$x_1 = x_0 - \left(\frac{1}{2} \arctan(x_0 - 2) \right) (2 + 2(x_0 - 2)^2) = 3.32 - \left(\frac{1}{2} \arctan(3.32 - 2) \right) (2 + 2(3.32 - 2)^2) \\ = 0.7902338003$$

$$x_2 = x_1 - \left(\frac{1}{2} \arctan(x_1 - 2) \right) (2 + 2(x_1 - 2)^2) = 0.7902338003 - \left(\frac{1}{2} \arctan(0.79[...]-2) \right) (2 + 2(0.79[...]-2)^2) \\ = 2.958246031$$

$$x_3 = \dots\dots\dots = 1.49256208884$$

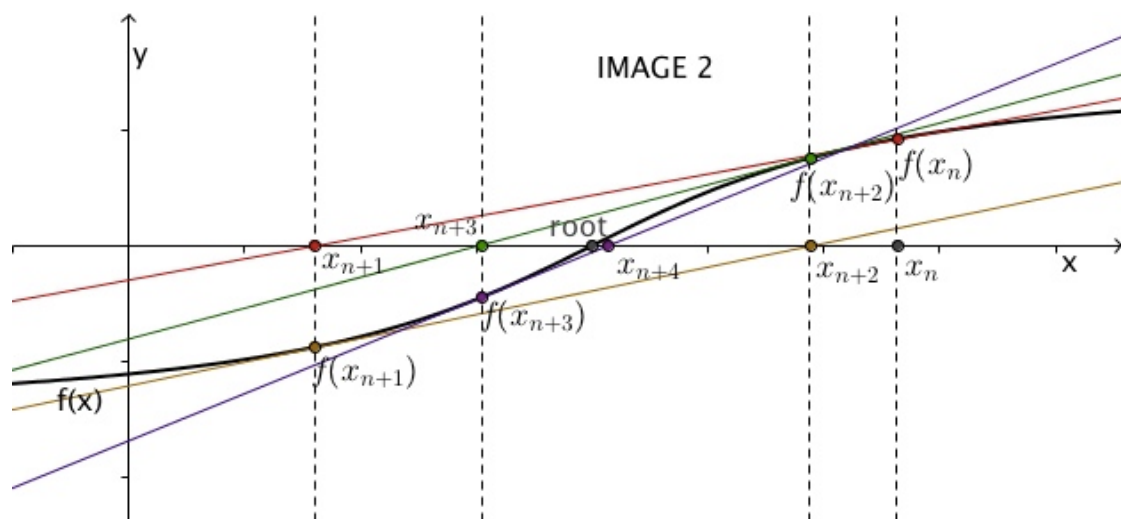
$$x_4 = \dots\dots\dots = 2.08305603519$$

$$x_5 = \dots\dots\dots = 1.99961856152$$

$$x_6 = \dots\dots\dots = 2.00000000004$$

$$x_7 = \dots\dots\dots = 2$$

A visualization (Image 2) of the process is presented below.



In this case, the iterative method's outcome may be compared to an algebraic solution to $f(x)$. This may be useful for examining the efficiency and speed of convergence when Newton's method is applied. Solving algebraically:

$$y = \frac{1}{2} \arctan(x - 2)$$

$$0 = \frac{1}{2} \arctan(x - 2)$$

$$0 = \arctan(x - 2)$$

$$\tan(0) = x - 2$$

$$0 = x - 2$$

$$x = 2$$

Hence, the iterative method does indeed converge to an existing root.

Another useful case study is a septic function. The Newton-Raphson method is, in this case, essential to deriving the function's solution because, according to the Abel's impossibility theorem, it is algebraically unsolvable as a septic function. Thus, let $g(x)$ be a real-valued septic function defined throughout its domain. Assume a root of this septic function is to be found. The derivation of the iterative algorithm for $g(x)$ may be a simpler task than for the transcendental function $f(x)$ investigated above, but the iteration itself is conceptually and graphically more scrambled. But to piece together the $g(x)$'s algorithm, its first derivative must firstly be computed:

$$g(x) = \frac{1}{10}(x - 2)^7 - (x - 3)^6 + 1$$

$$g'(x) = 1 \cdot \left(\frac{7}{10}(x - 2)^6\right) - 1 \cdot (6(x - 3)^5) = \frac{7}{10}(x - 2)^6 - 6(x - 3)^5$$

The function and its derivative may then be inserted into the general algorithm to create the function-specific algorithm for $g(x)$:

$$x_{n+1} = x_n - \frac{g(x_n)}{g'(x_n)}$$

$$x_{n+1} = x_n - \frac{\frac{1}{10}(x_n - 2)^7 - (x_n - 3)^6 + 1}{\frac{7}{10}(x_n - 2)^6 - 6(x_n - 3)^5}$$

Again, an initial approximation of an x -value in the vicinity of the sought root must be made to begin the iteration. The initial approximation, x_0 , made for $g(x)$ is $x=0.5$. This value is chosen for several reasons. In the first instance, this value satisfies the conditions for quadratic convergence which the Proof of Quadratic Convergence sets out (the reasons for which, as indicated earlier, shall not be outlined here). Yet an appropriate initial estimate may be deduced even with little familiarity with the proof. The general septic function of $g(x)$ has been translated horizontally to the right – to a value in the vicinity of $x=2$ and $x=3$. It has also been shifted upwards, as indicated by its constant term. Furthermore, because the function consists of the general form's septic and sextic terms only, and since it has been divided by 10, it has no stationary points. It thus maintains that inverted, rotated sigmoidal shape that is typical for odd-degree functions. From this, and the fact that the function's first term has been multiplied by one-tenth, it can be deduced that $g(x)$ has a very slim shape and must therefore intersect the x -axis once, at some value in the neighborhood of $x=2$ and $x=3$. It must

additionally be noted that at $x=0$ (the function's y-intercept), $g(x)$ has a value just below -740, which implies that an approximation must be made which is at a fair distance from the y-axis, since the $g(x)$ rapidly distance itself from $y=0$ (the location of any potential roots) as x approaches zero. With these considerations made, the initial approximation of $x=0.5$ can be expected to produce a reliable result when inserted into the root-finding algorithm:

$$x_{n+1} = x_n - \frac{\frac{1}{10}(x_n - 2)^7 - (x_n - 3)^6 + 1}{\frac{7}{10}(x_n - 2)^6 - 6(x_n - 3)^5}$$

$$x_0 = \frac{1}{2}$$

$$x_1 = x_0 - \frac{\frac{1}{10}(x_0 - 2)^7 - (x_0 - 3)^6 + 1}{\frac{7}{10}(x_0 - 2)^6 - 6(x_0 - 3)^5} = \frac{1}{2} - \frac{\frac{1}{10}(\frac{1}{2} - 2)^7 - (\frac{1}{2} - 3)^6 + 1}{\frac{7}{10}(\frac{1}{2} - 2)^6 - 6(\frac{1}{2} - 3)^5} = 0.912265885826$$

$$x_2 = x_1 - \frac{\frac{1}{10}(x_1 - 2)^7 - (x_1 - 3)^6 + 1}{\frac{7}{10}(x_1 - 2)^6 - 6(x_1 - 3)^5} = \frac{1}{2} - \frac{\frac{1}{10}(0.91[\dots] - 2)^7 - (0.91[\dots] - 3)^6 + 1}{\frac{7}{10}(0.91[\dots] - 2)^6 - 6(0.91[\dots] - 3)^5} = 1.25510615434$$

$$x_3 = x_2 - \frac{\frac{1}{10}(x_2 - 2)^7 - (x_2 - 3)^6 + 1}{\frac{7}{10}(x_2 - 2)^6 - 6(x_2 - 3)^5} = \dots = 1.53540359299$$

$$x_4 = \dots = 1.75474437895$$

$$x_5 = \dots = 1.90662432638$$

$$x_6 = \dots = 1.98219356914$$

$$x_7 = \dots = 1.9992391254$$

$$x_8 = \dots = 1.99999855524$$

$$x_9 = \dots = 1.99999999999$$

$$x_{10} = \dots = 2$$

Again the function's root is estimated to be approximately $x=2$.

Conditions and possibilities of failure:

The Newton-Raphson method is highly effective, although convergence is limited by several conditions and practical difficulties that are important to take into consideration. It must preliminarily be noted that the method depends on the ready obtainability of a function's derivative. Differentiability – and straightforward differentiability – are fundamental prerequisites to the method working efficiently, or working at all. If differentiation is highly complex, the iterative algorithm is only derived with difficulty and, once produced, is likely to be computationally unmanageable.

Newton's method will fail if its initial approximation is a stationary point. In this case, the linearization at the function under consideration is a horizontal constant with no x-intercept. As the method is based on repeatedly computing the x-intercepts of tangent lines to a function, a horizontal linearization within the iterative process will not lead to a productive outcome. Consider the function

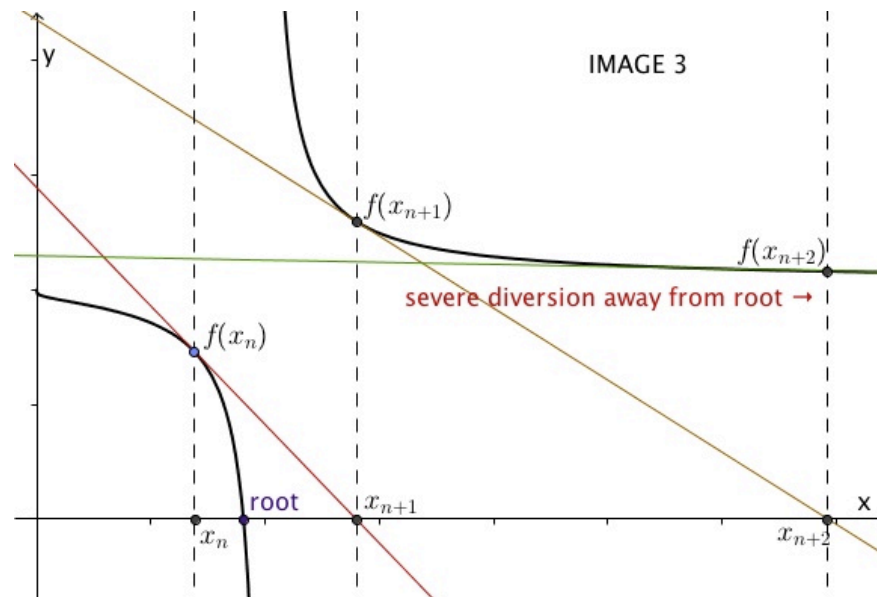
$y = (x-2)^2 - 2$, for instance. If the initial estimate is made at $x=2$, then the function will equal -2. This is the position of the parabola's minimum, however, and its linearization at this point will produce a horizontal constant which will cause the iterative method to fail. In this rather simplified case, a different initial estimate should be made.

Probably the main practical difficulty with Newton's method is its vulnerability to what is called "overshoot" (Weisstein). It describes the gradual divergence of the iterative approximations and the sought root. Initial estimates too far from the sought root, or which require linearizations to be made across stationary points, cause divergence because they may give rise to uncontrolled oscillations of the linearizations between the function and the x-axis (Weisstein). Furthermore, the iterative process may be drawn away from the sought root and toward stationary points that, in shape, might imitate a root (Weisstein).

Overshoot frequently occurs with rational functions, reciprocal transcendental functions, and oscillating functions. For oscillating functions, there is high risk of electing a stationary point as the initial estimate due to the high abundance of stationary points. Furthermore, basins of attraction around stationary points that imitate roots, but cause divergence from the sought root, are also likely to be abundant. (Basins of attraction are shapes of curves around which the iterations are likely to concentrate around.) Linearizing across asymptotes also frequently causes divergence. The part-functions on their respective sides of an asymptote may have inverted or entirely different shapes: thus basins of attractions typically change across asymptotes. The behavior of the algorithm when applied to the function

$$f(x) = \frac{1}{10 \ln(x)} + 1$$

is indicative of the rapidity with which the presence of asymptotes causes divergence. The graph shown below (Image 3) visualizes the extent of the divergence for $f(x)$.



As indicated in the method's description, most of the practical difficulties and failures may be circumvented by electing an initial estimate that satisfies the conditions set out by the Proof of Quadratic Convergence for Newton's Iterative Method. If this is the case, it is proven that the initial

approximation will invariably converge to the sought root, with at least quadratic rate of convergence. Thus, the careful selection of an initial approximation, under consideration of the conditions set out by the proof of quadratic convergence, is essential to the algorithm's successful application.

Derivation of optimization algorithm

The Newton-Raphson method for optimization seeks to build a sequence of increasingly accurate approximations, x_n , of a value of x for which $f'(x) = 0$ (stationary point). An initial estimate, x_0 , converges towards the values of x at which there is a stationary point ($x: f'(x)=0$).

The derivation of an optimization algorithm for Newton's method involves the reformulation of the function under consideration into a Taylor expansion or series (Smith). A Taylor series may be described as a representation or reformulation of a given function as an infinite sum of terms which consist of the values of the function's derivatives at a certain point. In simple terms, a finite Taylor series can be described as an approximation of the shape of a curve which, as the number of derivatives taken into account increases, becomes an increasingly accurate representation of the curve. If a function is differentiated infinitely, a Taylor series can be seen as its exact representation – the function and the series are equivalent. Consider, the Taylor series of an infinitely differentiable, real-valued function $f(x)$ at a certain x -value, a , is equivalent to the power series

$$f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots + \frac{f^n(a)}{n!}(x-a)^n$$

or in sigma notation for an infinitely differentiable function:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^n(a)}{n!}(x-a)^n$$

where $n!$ is the factorial of n , and $f^n(a)$ denotes the n th derivative of the function f at point $x=a$, which the series centers on. If not infinitely differentiated, a Taylor expansion is an approximation of the shape of a curve.

Suppose the second-order Taylor expansion of a function $f(x)$ is defined to be $f_2(x)$. Also suppose this expansion centers on the approximations of iteration, x_n , which gradually converge with the sought value of x at which a stationary point is to be found. In this case, the Taylor expansion reads:

$$f_2(x) = f(x_n) + \frac{f'(x_n)}{1!}(x-x_n) + \frac{f''(x_n)}{2!}(x-x_n)^2$$

$$f_2(x) = f(x_n) + f'(x_n)(x-x_n) + \frac{1}{2}f''(x_n)(x-x_n)^2$$

Now, for purposes of simplicity, let $(x - x_n) = \Delta x$. The Taylor expansion is now equivalent to:

$$f_2(x) = f(x_n) + f'(x_n)\Delta x + \frac{1}{2}f''(x_n)\Delta x^2$$

This representation itself has an extremum when its derivative with respect to Δx is equal to zero (and $f(x_n)$, $f'(x_n)$, and $f''(x_n)$ are assumed to be constant coefficients). This process of differentiation may be presented as

$$\frac{d}{d\Delta x}(f_2(x)) = f'(x_n) + f''(x_n)\Delta x$$

which, when set equal to zero, renders:

$$f'(x_n) + f''(x_n)\Delta x = 0$$

$$(\Delta x = x - x_n)$$

This equation may be rearranged to indicate a relation between the iteration estimates (x_n) derived and the first and second derivatives of the function under consideration.

$$f'(x_n) + f''(x_n)(x - x_n) = 0$$

$$x - x_n = -\frac{f'(x_n)}{f''(x_n)}$$

$$x = x_n - \frac{f'(x_n)}{f''(x_n)}$$

Therefore (provided that $f(x)$ is well-approximated by its Taylor series, it is twice differentiable, and any initial approximations chosen satisfy the conditions discussed earlier) the sequence of iteration estimates (x_n) defined as

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

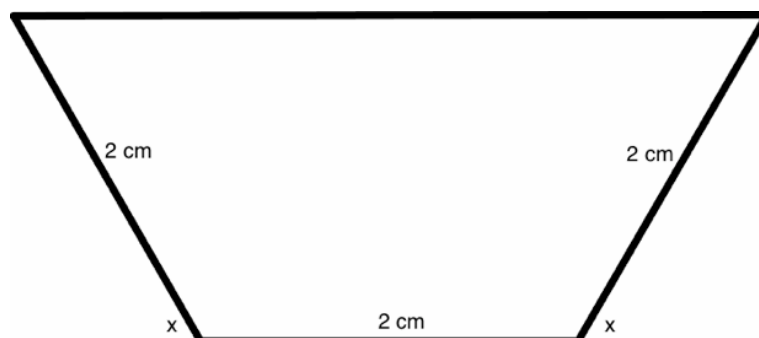
converges towards the value of x at which $f'(x)=0$ ($x: f'(x)=0$ where $f''(x_n) \neq 0$) (Smith). As here shown, the Newton-Raphson method of linear approximation, may also be reformulated into an algorithm of quadratic approximation for one-dimensional optimization (Smith). The conditions of convergence for the optimization algorithm are very similar to those for the root-finding method, but shall not be expanded on here. (The above shown derivation is based on information presented in Mark D. Smith's article for MIT.)

As already explained in the introduction, the significance of this algorithm is profound. It provides a method of optimization for systems of algebraically unsolvable equations, which often are the basis of mathematical models in the social and physical sciences. But to test the derived algorithm, a case study from the common field of application civil engineering shall be examined. Especially in descriptions or visualizations of planned designs, quantities of measurement may be related with each other through complicated transcendental, circular, or other non-linear functions. Yet accuracy in design, even to the nano-scale, is often vital to the implemented prototype's successful working. It is worth while demonstrating the application of the Newton-Raphson method for one-dimensional optimization in **an example borrowed from civil engineering**.

This example problem was taken from a video lecture by the University of Southern California. Suppose the cross-sectional area A of a trapezoidal gutter with a base length and edge length of 2cm is described by the equation:

$$A(x) = 4 \sin x (1 + \cos x)$$

One is to find the angle x which the sides of the gutter makes to the ground that maximizes the gutter's cross-sectional area.



The angle x is acute, the initial estimate should thus be less than 0.5π . The initial estimate made here is 0.25π .

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

$$x_{n+1} = x_n - \frac{4(\cos x_n + \cos^2 x_n - \sin^2 x_n)}{-4 \sin x_n (1 + 4 \cos x_n)} = x_n + \frac{(\cos x_n + \cos^2 x_n - \sin^2 x_n)}{\sin x_n (1 + 4 \cos x_n)}$$

$$x_0 = \frac{\pi}{4}$$

$$x_1 = x_0 + \frac{(\cos x_0 + \cos^2 x_0 - \sin^2 x_0)}{\sin x_0 (1 + 4 \cos x_0)} = \frac{\pi}{4} + \frac{(\cos \frac{\pi}{4} + \cos^2 \frac{\pi}{4} - \sin^2 \frac{\pi}{4})}{\sin \frac{\pi}{4} (1 + 4 \cos \frac{\pi}{4})} = 1.046602038$$

$$x_2 = x_1 + \frac{(\cos x_1 + \cos^2 x_1 - \sin^2 x_1)}{\sin x_1 (1 + 4 \cos x_1)} = \frac{\pi}{4} + \frac{(\cos(1.04[...]) + \cos^2(1.04[...]) - \sin^2(1.04[...]))}{\sin(1.04[...])(1 + 4 \cos(1.04[...]))} = 1.047197449$$

$$x_3 = \dots\dots\dots = 1.047197551$$

$$x_4 = \dots\dots\dots = 1.047197551\dots$$

The stationary point found must be classified with the second derivative test if maximization is to be performed:

$$f''(1.047197551) = -4(\sin(1.047197551))(1 + 4 \cos(1.047197551)) = -10.4$$

$$\Rightarrow f''(x) < 0$$

$\therefore$ local max

$$1.047197551 = 60^\circ = \frac{\pi}{3}$$

The angle at which the gutter's cross sectional area is maximized is thus deduced. It may now be inserted into the original relation and the maximum area computed:

$$f\left(\frac{\pi}{3}\right) = 4\left(\sin\left(\frac{\pi}{3}\right)\right)\left(1 + \cos\left(\frac{\pi}{3}\right)\right) = 4 \cdot \left(\frac{\sqrt{3}}{2}\right) \cdot \left(\frac{3}{2}\right) = 3\sqrt{3} \text{ cm}^2$$

Conclusion

The Newton-Raphson method's conceptual simplicity is very much the root of its genius. A function is linearized at an initial approximation of its root and the x-intercept of this tangent line serves as the new, closer estimate. This process is repeated to give rise to an iterative method for the approximation of the roots of real-valued, non-linear equations at a quadratic rate of convergence. This method may then be generalized into the complex plane, quadratic approximation, multi-dimensional optimization, and solving transcendental equations. From a philosophical perspective, Newton's method may very well constitute mathematical beauty. Not necessarily in result, but in proof and method Newton's algorithm is profoundly elegant; it combines tremendous versatility with rare succinctness and accessibility. It thus constitutes an interesting philosophical study as it satisfies and challenges many of the conditions attributed to mathematical beauty, a rather abstract and counter-intuitive concept in itself. On one hand, it may be unreliable and constrained by context; but on the other its elegance and conceptual ingenuity – the extent to which it unifies utility and clarity – make it a profoundly beautiful method.

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